Extracranial metastasis of glioblastoma with genomic analysis: a case report and review of the literature.

PMID: 36093544 | Indexed in PubMed

Glioblastoma (GBM) is the first most frequent type of primary malignant brain tumors in adults. It is basically confined to the brain, and extracranial metastases (ECM) are rare. The genomic features of GBM with ECM are not fully elucidated. Here, we present a case of a male patient with headache and left eye vision loss for 2 months who had a left occipital lobe tumor. GBM of grade IV [isocitrate dehydrogenase 1 (IDH-1) wild type] was diagnosed based on the histological profiles of intracranial tumor according to the World Health Organization standard. ECM of GBM located in the mediastinal lymph node occurred 6 months after resection of the intracranial tumor. High throughput gene sequencing was performed using ECM lesions. Mutated genes included tumor protein 53 (TP53), CUB and Sushi multiple domains 3 (CSMD3), poly(ADP-ribose) polymerase family member 4 (PARP4), and PTEN. The patient underwent surgery, radiotherapy, chemotherapy, and anti-angiogenic drug treatment. Unfortunately, the patient died 8 months after surgery. ECM of GBM is rare, and its prognosis is very poor. Mutated genes in ECM included TP53, CSMD3, PARP4, and PTEN in our case. Genomic analysis provides important insights into GBM and its ECM.

Bone Metastases of Glioblastoma: A Case Report and Review of the Literature.

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Glioblastoma (GBM) is the most common primary intracranial tumor and originates from the small pool of adult neural stem and progenitor cells (NSPCs). According to the World Health Organization (WHO) classification of brain tumors, gliomas are classified into grades I-IV, and GBM is defined as the highest grade (IV). GBM can be disseminated by cerebrospinal fluid (CSF), but extracranial metastasis is rare. Additionally, the pathway and mechanism involved remain unclear. We report a rare case of left temporal lobe GBM with multiple bone metastases and soft tissue metastasis. This 49-year-old right-handed man who was diagnosed with GBM underwent surgery on May 9, 2017, followed by radiochemotherapy in June 2017. On August 13, 2019, local relapse was found. Then, the patient received a second surgery but not radiochemotherapy. In November 2019, the patient was reported to be suffering from low back pain for nearly 1 month. On December 6, 2019, magnetic resonance imaging (MRI) of the thoracolumbar vertebrae and abdominal computed tomography (CT) confirmed metastases on the ninth posterior rib on the right, the third anterior rib on the left, and the T7 and T10 vertebrae and their appendages. CT-guided rib space-occupying puncture biopsy was performed, and GBM was identified by pathology. We should pay attention to extracranial metastasis of GBM. Timely detection and early treatment improve overall quality of patients' life. The extracranial metastasis in this patient may have occurred through the spinal nerve root or intercostal nerve. Further clinical observations are required to clarify the pathway and mechanism involved.

[Glioblastoma multiforme with liver metastasis--case report].

PMID: 7546923 | Indexed in PubMed

Extracranial metastasis of glioblastoma is rare. This is an autopsy case report of a patient with glioblastoma multiforme found to have metastasized to the liver. A 42-year-old woman was admitted with a chief complaint of headache. Physical and neurological examinations on admission showed no abnormalities. CT and MRI demonstrated a tumor in the left parietooccipital region with invasion into the subependymal area of the left lateral ventricular trig-one. A cerebral angiogram showed tumor staining in the same area. Subtotal tumor resection was performed uneventfully. The microscopic diagnosis was glioblastoma multiforme. Postoperatively, the patient underwent whole brain and local irradiation, and intra-arterial ACNU infusion therapy. One month later, she developed low back pain, probably due to spinal dissemination. Postmortem examination showed local recurrence of the tumor and subarachnoidal dissemination not only in the base of the skull but in the lower spinal cord. Tumor was also observed in the liver, but no lung or lymph node metastasis was detected. Metastasis to the liver in this patient is believed to have occurred via the anastomosis between the vertebral and portal venous system.

Glioblastoma multiforme metastases to the masticator muscles and the scalp.

PMID: 29685418 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most common malignant primary brain tumor. Metastases outside the nervous system are a rare phenomenon. We present a pictographically striking case of GBM with metastases in the masticatory muscles of the infratemporal fossa and the scalp, in the context of a dense intracranial dissemination. Extracranial metastases of GBM have been reported anecdotally, while, to our knowledge, masticatory muscle metastases have been previously reported in only one case.

Cervical metastatic glioblastoma multiforme.

PMID: 23374526 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most common and most malignant primary brain tumour in adults. In spite of the hostile nature of glioblastoma multiforme (GBM), extracranial spread is not a common event. With improving management choices and survival times, reports of extracranial occurrence of GBM have increased. Most commonly these metastases are to the lungs, lymph nodes, neck, skull, scalp, liver, and bones; may be evident on routine follow-up images of the original lesion. Head and neck metastasis of GBM can be debilitating. We present a case of cervical metastasis of GBM and discuss possible mechanisms of extraneural spread of this tumour.

Multiple extracranial metastases from secondary glioblastoma multiforme: a case report and review of the literature.

PMID: 19898745 | Indexed in PubMed

Extracranial metastasis of glioblastoma multiforme (GBM) is very rare, in spite of very aggressive tumor behavior and being documented in only a few patients. In this article we present a 25-year-old man with secondary glioblastoma associated with extracranial progression and distant metastasis. He was diagnosed by magnetic resononce (MR) with an intracranial lesion in the right parietofrontal region, which was subsequently resected. Histology revealed a diffuse astrocytoma (grade II). The tumor recurred 1 year later and the patient received a second craniotomy. A diagnosis of GBM was made. After radiotherapy, he presented with right cervical lymph node metastases. The cytomorphological features supported a diagnosis of metastatic glioblastoma multiforme. The neck dissection was made and histology confirmed the fine needle aspiration diagnosis of glioblastoma multiforme. MR with diffusion-weighted imaging revealed right cervical lymph node metastases and multi-bone metastases (mainly pelvic bone) 3 weeks later.

Bone marrow metastases from glioblastoma multiforme--A case report and review of the literature.

PMID: 15925996 | Indexed in PubMed

Clinically detected extra-cranial metastases from glioblastoma multiforme (GBM) are quite rare, with an incidence of <2% reported in the published literature. Among the various reported sites of systemic metastases from GBM, there are few cases of clinically symptomatic bone marrow metastasis. The case of a patient developing systemic dissemination of a GBM is described. A 60-year-old man with GBM who developed back pain, thrombocytopenia and subsequently neurological deficits was found to have extensive bony and bone marrow metastases. Previously reported cases of extra-cranial systemic spread of GBM and attempts made in the literature to explain the possible routes of extra-neural dissemination are reviewed.

Extracranial oral cavity metastasis from glioblastoma multiforme: A case report.

PMID: 27699039 | Indexed in PubMed

Glioblastoma multiforme is the most common primary malignant brain tumor. The clinical outcome following diagnosis remains extremely poor. The treatment of choice is wide surgical resection of the visible tumor, frequently followed by adjuvant combined radiochemotherapy (RCTx) with temozolomide as the chemotherapeutic agent. Extracranial metastases are extremely rare, with <200 cases of extracranial metastases from glioblastoma multiforme reported in the literature to date. We herein present a case of a patient suffering from a fast-growing metastasis to the oral cavity, completely filling the buccal cavity within 2 weeks, as the only manifestation of recurrent glioblastoma multiforme following initial surgical resection and adjuvant RCTx.

Primary presentation of glioblastoma multiforme with leptomeningeal metastasis in the absence of previous craniotomy: a case report.

PMID: 15577447 | Indexed in PubMed

Glioblastoma multiforme is a highly malignant glioma with a well-known tendency for intracranial spread but rarely for extracranial spread. We report a case of an adult woman who presented with symptoms of leptomeningeal metastasis from an intracranial glioblastoma multiforme located adjacent to the lateral ventricle. There have been very few cases of glioblastoma multiforme presenting with leptomeningeal metastasis in the absence of previous therapeutic intervention.

Scalp Metastasis from Glioblastoma Multiforme: A Case Report and Literature Review.

PMID: 28182828 | Indexed in PubMed

Glioblastoma multiforme (GBM) is a common malignant brain tumor that rarely metastasizes extracranially, despite its aggressive clinical course. This report details the case of a young man presenting with a single subcutaneous localization of GBM that arose six months after initial surgery and recurred after excision. Only six other cases of scalp metastasis of GBM following surgery have been described in the literature, each with peculiar features. Whenever feasible, surgery is the most effective way to obtain local control of disease. However, a correct approach must be carefully planned to minimize the risks of recurrence and wound dehiscence.